

PHYSICS

TARGET : JEE- Advanced

CAPS-9

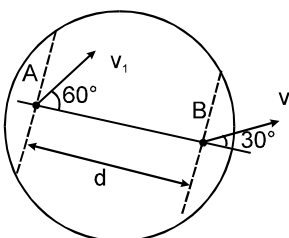
ROTATIONAL DYNAMICS-2

SCQ (Single Correct Type) :

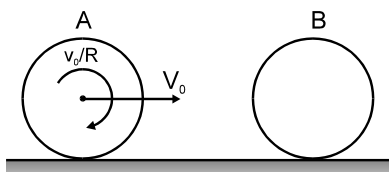
1. A solid sphere of mass m and radius r is gently placed on a conveyer belt moving with constant velocity V . If the coefficient of friction between the belt and sphere is $\frac{2}{7}$, the distance travelled by the centre of the sphere before it starts pure rolling is



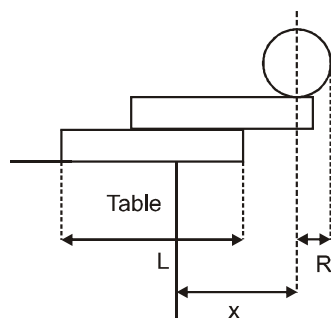
- (A) $\frac{V^2}{7g}$ (B) $\frac{2V^2}{49g}$ (C) $\frac{2V^2}{5g}$ (D) $\frac{2V^2}{7g}$
2. Two points A & B on a disc have velocities v_1 & v_2 at some moment. Their directions make angles 60° and 30° respectively with the line of separation as shown in figure. The angular velocity of disc is :



- (A) $\frac{\sqrt{3}v_1}{d}$ (B) $\frac{v_2}{\sqrt{3}d}$ (C) $\frac{v_2 - v_1}{d}$ (D) $\frac{v_2}{d}$
3. A hollow smooth uniform sphere A of mass ' m ' rolls without sliding on a smooth horizontal surface. It collides elastically and head on with another stationary smooth solid sphere B of the same mass m and same radius. The ratio of kinetic energy of 'B' to that of 'A' just after the collision is :



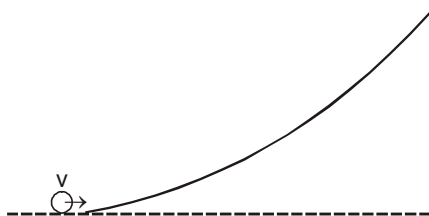
- (A) 5 : 2 (B) 1 : 1 (C) 2 : 3 (D) 3 : 2
4. Two identical uniform rectangular blocks (with longest side L) and a solid sphere of radius R are to be balanced at the edge of a heavy table such that the centre of the sphere remains at the maximum possible horizontal distance from the vertical edge of the table without toppling as indicated in the figure.



If the mass of each block is M and of the sphere is $M/2$, then the maximum distance x that can be achieved is

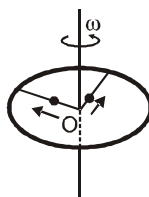
- (A) $8L/15$ (B) $5L/6$ (C) $(3L/4 + R)$ (D) $(7L/15 + R)$

5. A small object of uniform density rolls up a curved surface with an initial velocity v . It reaches up to a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is



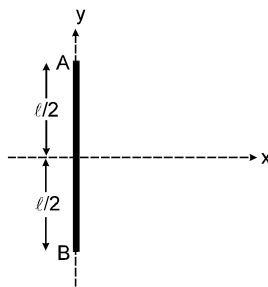
- (A) ring (B) solid sphere (C) hollow sphere (D) disc

6. A ring of mass M and radius R is rotating with angular speed ω about a fixed vertical axis passing through its centre O with two point masses each of mass $\frac{M}{8}$ at rest at O . These masses can move radially outwards along two massless rods fixed on the ring as shown in the figure. At some instant the angular speed of the system is $\frac{8}{9}\omega$ and one of the masses is at a distance of $\frac{3}{5}R$ from O . At this instant the distance of the other mass from O is :



- (A) $\frac{2}{3}R$ (B) $\frac{1}{3}R$ (C) $\frac{3}{5}R$ (D) $\frac{4}{5}R$

7. A uniform rod of mass m , length ℓ is placed over a smooth horizontal surface along y -axis and is at rest as shown in figure. An impulsive force F is applied for a small time Δt along x -direction at point A after this rod moves freely. The x -coordinate of end A of the rod when the rod becomes parallel to x -axis for the first time is (initially the coordinate of centre of mass of the rod is $(0, 0)$) :



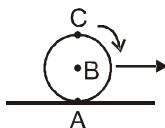
- (A) $\frac{\pi l}{12}$ (B) $\frac{l}{2} \left(1 + \frac{\pi}{12} \right)$ (C) $\frac{l}{2} \left(1 - \frac{\pi}{6} \right)$ (D) $\frac{l}{2} \left(1 + \frac{\pi}{6} \right)$

MCQ (One or more than one correct) :

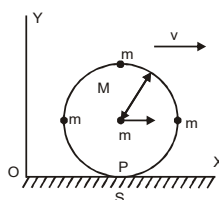
8. A rigid body undergoing pure rolling encounters horizontal rigid tracks AB and BC as shown. AB is smooth surface and BC is rough surface with $\mu = 1$. Which of the following statements is/are **correct** :



- (A) Angular momentum of the rigid body is conserved only about a point on the horizontal surface.
 (B) Angular momentum of the rigid body is conserved about every point in space.
 (C) In part BC, there will be no frictional force on the rigid body.
 (D) In part BC, frictional force will act opposite to velocity of rigid body.
9. A sphere is rolling without slipping on a fixed horizontal plane surface. In the figure, A is the point of contact, B is the centre of the sphere and C is its topmost point. Then,



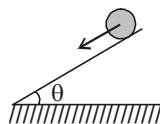
- (A) $\vec{V}_C - \vec{V}_A = 2(\vec{V}_B - \vec{V}_C)$ (B) $\vec{V}_C - \vec{V}_B = \vec{V}_B - \vec{V}_A$
 (C) $|\vec{V}_C - \vec{V}_A| = 2 |\vec{V}_B - \vec{V}_C|$ (D) $|\vec{V}_C - \vec{V}_A| = 4 |\vec{V}_B|$
10. A uniform circular disc of mass M. and radius R has four particles (each of mass m) rigidly attached to it. The disc rolls on the surface S without slipping. At the instant shown the centre of the disc has velocity v_0 ,



- (A) The kinetic energy of the disc-particle assembly is $\left(\frac{9m}{2} + \frac{3M}{4} \right) v_0^2$



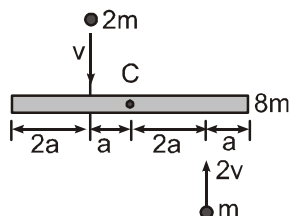
- (B) The velocity of the centre of mass of the assembly is $\left(\frac{M+5m}{M+4m}\right)v_0\hat{i}$
- (C) The angular momentum of the assembly about the point of contact P is $\frac{3}{2}(M+6m)(Rv_0(-\hat{k}))$
- (D) The maximum speed of a point on the disc with respect to S is $2v_0$
11. A solid sphere is in pure rolling motion on an inclined surface having inclination θ .



- (A) frictional force acting on sphere is $f = \mu mg \cos \theta$.
- (B) f is dissipative force.
- (C) friction will increase its angular velocity and decreases its linear velocity.
- (D) If θ decreases, friction will decrease.

Comprehension Type Question:

A uniform bar of length $6a$ & mass $8m$ lies on a smooth horizontal table. Two point masses m & $2m$ moving in the same horizontal plane with speeds $2v$ and v respectively strike the bar as shown & stick to the bar after collision.



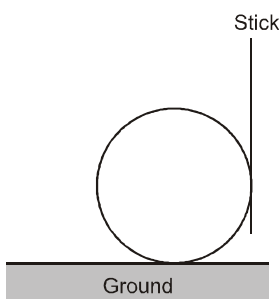
12. Velocity of the centre of mass of the system is
- (A) $\frac{v}{2}$ (B) v (C) $\frac{2v}{3}$ (D) Zero
13. Angular velocity of the rod about centre of mass of the system is
- (A) $\frac{v}{5a}$ (B) $\frac{v}{15a}$ (C) $\frac{v}{3a}$ (D) $\frac{v}{10a}$
14. Total kinetic energy of the system, just after the collision is
- (A) $\frac{3}{5}mv^2$ (B) $\frac{3}{25}mv^2$ (C) $\frac{3}{15}mv^2$ (D) $3mv^2$

Numerical based Questions :

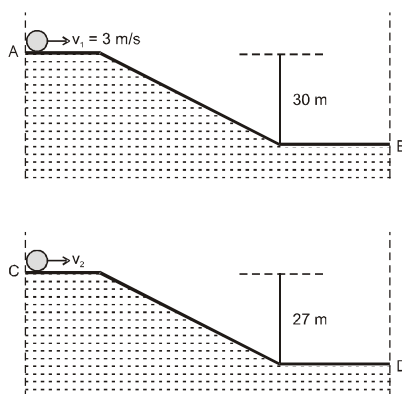
15. A solid billiard ball of radius ' R ' and mass ' m ' initially at rest is given a sharp impulse by a cue, held horizontally at a distance $\frac{R}{2}$ above the centre. Just after the impulse, the velocity of centre of mass of the ball is $v = 10$ m/s. The coefficient of friction between ball and table is

$\mu = \frac{1}{2}$. The ball starts rolling without slipping t seconds after impulse is given. Find value of $\frac{1}{t}$ (in sec^{-1})

16. A boy is pushing a ring of mass 2 kg and radius 0.5 m with a vertical stick as shown in the figure. The stick applies a normal force of 2 N on the ring and rolls it without slipping with an acceleration of 0.3 m/s^2 . The coefficient of friction between the ground and the ring is large enough that rolling always occurs and the coefficient of friction between the stick and the ring is $(P/10)$. The value of P is



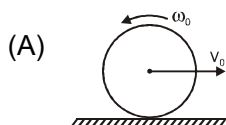
17. Two identical uniform discs roll without slipping on two different surfaces AB and CD (see figure) starting at A and C with linear speeds v_1 and v_2 , respectively, and always remain in contact with the surfaces. If they reach B and D with the same linear speed and $v_1 = 3 \text{ m/s}$, then v_2 in m/s is ($g = 10 \text{ m/s}^2$)



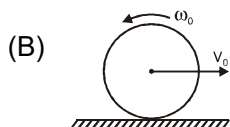
Matrix Match Type :

18. A uniform disc of mass M and radius R lies on a fixed rough horizontal surface at time $t = 0$. Initial angular velocity ω_0 of each disc (magnitude and sense of rotation) and horizontal velocity v_0 of centre of mass is shown for each situation of column-I. Match each situation in column-I with the results given in column-II.

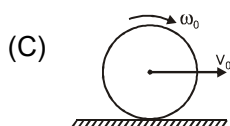
Column-I



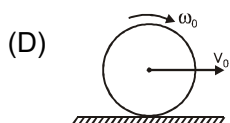
It is given that $v_0 = 2R\omega_0$



It is given that $2v_0 = R\omega_0$



It is given that $v_0 = 2R\omega_0$



It is given that $2v_0 = R\omega_0$

Column-II

(p) The angular speed keeps on decreasing

till the disc stops slipping.

(q) After the disc stops slipping, the angular

velocity is nonzero and in clockwise sense

(r) After the disc stops slipping, the velocity

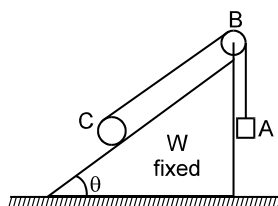
of centre of disc is towards right

(s) After the disc stops slipping, the

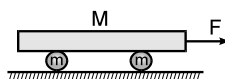
kinetic energy of disc is less than its initial value.

Subjective Type Questions :

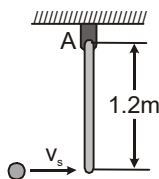
19. In the figure shown there is a fixed wedge W of inclination ' θ '. A is a block, B is a disc and ' C ' is a solid cylinder. A , B and C each has mass m . Assuming there is no sliding anywhere and string to be of negligible mass find :



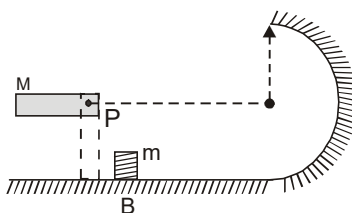
- (a) the speed of A after descending distance ' x '.
- (b) the friction force acting on the cylinder due to the wedge.
20. Two cylindrical rollers each of mass m are used to transport a log of mass M . If a horizontal force F acts on the log of wood find its acceleration. There is no slipping anywhere



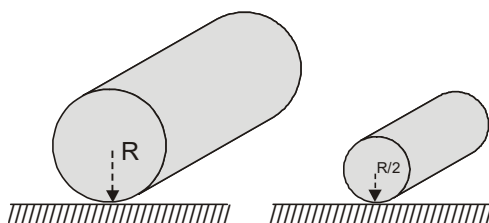
21. A 2 kg sphere moving horizontally to the right with an initial velocity of 5 m/s strikes the lower end of an 8 kg rigid rod AB. The rod is suspended from a hinge at A and is initially at rest. Knowing that the coefficient of restitution between the rod and sphere is 0.80, determine the angular velocity of the rod and the velocity of the sphere immediately after the impact.



22. A rod of length R and mass M is free to rotate about a horizontal axis passing through hinge P as in figure. First it is taken aside such that it becomes horizontal and then released. At the lowest point the rod hits the small block B of mass m and stops. Find the ratio of masses such that the block B completes the circular track of radius R . Neglect any friction.



23. A plank of mass m_1 with a uniform sphere of mass m_2 placed on it rests on a smooth horizontal plane. A constant horizontal force F is applied to the plank. With what accelerations will the plank and the centre of the sphere move provided there is no sliding between the plank and the sphere ?
24. A carpet of mass ' M ' made of inextensible material is rolled along its length in the form of a cylinder of radius ' R ' and is kept on a rough floor. The carpet starts unrolling without sliding on the floor when a negligibly small push is given to it. Calculate the horizontal velocity of the axis of the cylindrical part of the carpet when its radius reduces to $R/2$.



25. The surface mass density (mass/area) of a circular disc of radius ' R ' depends on the distance x from the centre given as, $\sigma(x) = \alpha + \beta x$. Where α and β are positive constant find its moment of inertia about the line perpendicular to the plane of the disc through its centre.

ANSWER KEY (CAPS-9)

1. (A) 2. (D) 3. (D) 4. (A) 5. (D)
 6. (D) 7. (D) 8. (BC) 9. (BC)
 10. (ABCD) 11. (CD) 12. (D) 13. (A) 14. (A)
 15. 7 16. 4 17. 7
 18. (A) q,r,s (B) p,s (C) q,r,s (d) p,q,r,s
 19. **Method - I** (a) $V_A = \sqrt{\frac{8gx}{15}(2 - \sin\theta)}$ (b) $f = \frac{mg}{15}(1 + 7 \sin\theta)$
Method - II (a) $v = \sqrt{\frac{8g(2 - \sin\theta)x}{15}}$ (b) $f = \frac{mg}{15}(1 + 7 \sin\theta)$
 20. $4F/(4M + 3m)$ 21. $\omega = \frac{45}{14} = 3.21 \text{ rad/s (ccw)}, v_s = \frac{1}{7} = 0.143 \text{ m/s} \leftarrow$
 22. $\frac{M}{m} = \sqrt{15}$ 23. $w_1 = F/(m_1 + 2/7m_2); w_2 = 2/7 w_1.$ 24. $v = \sqrt{\frac{14gR}{3}}$
 25. $2\pi \left(\frac{\alpha R^4}{4} + \frac{\beta R^5}{5} \right)$